CH 8 — Taylor Polynomial and Taylor Theorem

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Recall: For a single variable function, $f(x)$, the second-order Taylor Polynomial at $x = a$ is

$$P_{2,a}(x) = f(a) + f'(a)(x-a) + \frac{1}{2}f''(a)(x-a)^2 = L_a(x) + \frac{1}{2}f''(a)(x-a)^2$$

This is the 2nd order polynomial which agrees with $f(x)$ at $x = a$ up to the 2nd order derivative, i.e.

$$P_{2,a}(a) = f(a)$$
$$P'_{2,a} = f'(a)$$
$$P''(2,a)(a) = f''(a)$$

📘 **Definition — Taylor Polynomial**

If $f(x,y) \in C^2$ at (a,b), then second-order Taylor Polynomial is defined of f at (a,b) by

$$P_{2,(a,b)}(a,b) = L_{(a,b)}(x,y) + \frac{1}{2}\big(f_{xx}(x-a)^2 + 2f_{xy}(a,b)(x-a)(y-b) + f_{yy}(y-b)^2\big)$$

📘 **Definition — Hessian Matrix**

The **Hessian Matrix** of $f(x,y)$ is

$$Hf(x,y) = \begin{bmatrix} f_{xx}(x,y) & f_{xy}(x,y) \\ f_{yx}(x,y) & f_{yy}(x,y) \end{bmatrix}$$

Examples:

1. Use both linearization and the second order Taylor Polynomial of $f(x,y) = e^{x+y}$ at $(0,0)$ to approximate $f(0.1, 0.2)$

   $L_{(0,0)}(0.1,0.2) = 1 + x + y\Big|_{(x,y)=(0.1,0.2)} = 1.3$

   $P_{2,(0,0)} = L_{(0,0)}(0.1,0.2) + \frac{1}{2}((x)^2 + 2(x)(y) + (y)^2)\Big|_{(x,y)=(0.1,0.2)}$

   $= 1.3 + \frac{1}{2}((0.1)^2 + 2(0.1)(0.2) + (0.2)^2) = 1.3 + (0.005 + 0.02 + 0.02) = 1.345$

2. Use the linearizatio and second-order Taylor polynomial of $f(x,y) = \frac{1}{2}x - y + x^5$ at $(0,0)$ to approximate $f(0.1, 0.1)$

   $L_{0,0}(x,y) = 0 + \big(\frac{1}{2} + 5(0)^4\big)(x) + (-1)(y) = \frac{1}{2}x - y$

   $L_{0,0}(0.1,0.1) = \frac{1}{2}(0.1) - 0.1 = -0.05$

   $P_{2,(0,0)}(x,y) = -0.05 + \frac{1}{2}[(20(0)^3)(x)^2 + 2(0)(x)(y) + (0)(y)^2] = -0.05$

   $P_{2,(0,0)} = -0.05$

In practical application, we don't know the value of what we are trying to approximate, so we cannot calculate the approximate the approximation error exactly.

Theorem — Taylor's Polynomial 1D

If $f''(x)$ exists on $[a, x]$, then $\exists a \in \mathbb{R}$ s.t $c \in (a, x)$ and

$$f(x) = f(a) + f'(a)(x-a) + R_{1,a}(x)$$

where $R_{1,a}(x) = \frac{1}{2}f''(c)(x-a)^2$

Theorem — Taylor's Polynomial 2D

If $f(x, y) \in C^2$ in some neighbourhood $N(a, b)$ of (a, b), then $\forall (a, b) \in N(a, b), \exists (c, d)$ on the line segment joining (a, b) and (x, y) s.t.

$$f(x, y) = f(a, b) + f_x(a, b)(x-a) + f)y(a, b)(y-b) + R_{1,(a,b)}(x, y)$$

where

$$R_{1,(a,b)}(x, y) = \frac{1}{2}\left[f_{xx}(c, d)(x-a)^2 + 2f_{xy}(c, d)(x-a)(y-b) + f_{yy}(c, d)(y-b)^2\right]$$

Proof

Consider the line segment joining (x, y) and (a, b). Consider $f(x, y)$ and let $g(t) = f\left(\vec{L}(t)\right)$.

Notice for a fixed (x, y), $x - a$ and $y - b$ are constant, let $x - a = k, y - b = h$.

$g(t) = f\left(\vec{L}(t)\right) = f(a + tk, b + th), 0 \leq t \leq 1$

$g'(t) = f_x\left(\vec{L}(t)\right)k + f_y\left(\vec{L}(t)\right)h$

$g''(t) = f_{xx}\left(\vec{L}(t)\right)k^2 + f_{xy}\left(\vec{L}(t)\right)kh + f_{xy}\left(\vec{L}(t)\right)hk + f_{yy}\left(\vec{L}(t)\right)h^2$

Since Taylor's theorem requires $f \in C^2$, then $f_{xy} = f_{yx}$

$$g''(t) = f_{xx}\left(\vec{L}(t)\right)k^2 + 2f_{xy}\left(\vec{L}(t)\right)kh + f_{yy}\left(\vec{L}(t)\right)h^2$$

We have $g''(t)$ is defined on $[0, 1]$, we can apply Taylor's theorem to $g(t)$ on $[0, 1]$, that is $\exists \tilde{c} \in (0, 1)$ s.t. for $t \in [0, 1]$, $g(t) = g(0) + g'(0)(t-0) + \frac{1}{2}g''(\tilde{c})(t-0)^2$

Let $t = 1, g(1) = g(0) + g'(0) + \frac{1}{2}g''(\tilde{c})$

$g(1) = f_x\left(\vec{L}(1)\right)k = f(x, y)$

$g(0) = f(a, b)$

$g'(0) = f_x(a, b)(x-a) + f_y(a, b)(y-b)$

Let $(c, d) = L(\tilde{c})$ point on the line segment

$\implies g''(\tilde{c}) = f_{xx}(c, d)(x-a)^2 + 2f_{xy}(c, d)(x-a)(y-b) + f_{yy}(a, b)(y-b)^2$

$f(x, y) = f(a, b) + f_x(a, b)(x-a) + f_y(a, b)(y-b) +$

$\frac{1}{2}f_{xx}(c, d)(x-a)^2 + 2f_{xy}(c, d)(x-a)(y-b) + f_{yy}(a, b)(y-b)^2$

□

Example:

Show that $x \geq 0, y \geq 0$ and $f(x,y) = \ln(1+x+2y)$ $R_{1,(0,0)} \leq 3(x^2+y^2)$

$f_x = \frac{1}{1+x+2y}, f_y = \frac{2}{1+x+2y}, f_{xx} = -\frac{1}{(1+x+2y)^2}, f_{xy} = f_{yx} = -\frac{2}{(1+x+2y)^2}, f_{yy} = -\frac{4}{(1+x+2y)^2}$

By Taylor theorem, $\exists(c,d)$ on the line joining $(0,0)$ and (x,y) s.t.

$$\begin{aligned} |R_{1,(0,0)}(x,y)| &= |\frac{1}{2}\big[f_{xx}(c,d)x^2 + 2f_{xy}(c,d)\big)xy + f_{yy}(c,d)y^2]| \\ &\leq \frac{1}{2}\big[|f_{xx}(c,d)|x^2 + 2|f_{xy}(c,d)|xy + |f_{yy}(c,d)|y^2\big] \end{aligned}$$

$|f_{xx}(c,d)| = \frac{1}{(1+c+2d)^2} \leq 1, |f_{xy}(c,d)| = \frac{2}{(1+c+2d)^2} \leq 2, |f_{yy}(c,d)| = \frac{4}{(1+c+2d)^2} \leq 4$

$\therefore |R_{1,(0,0)}(x,y)| \leq \frac{1}{2}[x^2 + 4|x\|y| + 4y^2] \overset{2|x\|y| \leq x^2+y^2}{=} \frac{1}{2}(x^2 + 2(x^2+y^2) + 4y^2)$

$= \frac{3}{2}x^2 + 3y^2 \leq 3(x^2+y^2) = 3\|(x,y) - (0,0)\|^2$

We can always have an upper bound that is greater than the current upper bound.

In general we get the below theorem.

> 📐 **Theorem — Corollary of Upper Bound of 2D Taylor Polynomial**
> If $f(x,y) \in C^2$ in some neighbourhood $N(a,b)$ of (a,b), then $\exists M \in \mathbb{R}_{\geq 0}$ s.t.
>
> $$R_{1,(a,b)}(x,y) \leq M\|(x,y) - (a,b)\|^2 \quad \forall (x,y) \in N(a,b)$$

> 💡 **Tip — Taylor Theorem usage**
>
> 1. Always use the triangle inequality first to obtain the error as a sum of non-negative terms
> 2. DO NOT substitute (a,b) into the second partial derivatives directly, instead use an arbitrary (c,d) as unknown point

Examples:

Let $f(x,y) = e^{-2x+y}$ Show that the error in $L_{1,1}$ is at most $3e\big[(x-1)^2 + (y-1)^2\big]$
if $0 \leq x \leq 1, 0 \leq y \leq 1$

$f_x = -2e^{-2x+y}, f_y = e^{-2x+y}; f_{xx} = 4e^{-2x+y}, f_{xy} = f_{yx} = -2e^{-2x+y}, f_{yy} = e^{-2x+y}$

$c,d = (0,1)$

$|f_{xx}(0,1)| \leq 4e^{-2(0)+1} = 4e, |f_{xy}(0,1) \leq 2e^{-2(0)+1}| = 2e, |f_{yy}(0,1)| \leq e^{-2(0)+1} = e$

By Taylor's theorem, $\exists(c,d)$ between $(1,1)$ and (x,y) s.t

$$
\begin{aligned}
R_{1,(1,1)}(x,y) &= \frac{1}{2}|f_{xx}(c,d)(x-1)^2 + 2f_{xy}(c,d)(x-1)(y-1) + f_{yy}(c,d)(y-1)^2| \\
&\leq \frac{1}{2}\big[|f_{xx}(c,d)|(x-1)^2 + 2|f_{xy}(c,d)(x-1)(y-1)| + |f_{yy}(c,d)|(y-1)^2\big] \\
&\leq \frac{1}{2}[4e(x-1)^2 + 4e|x-1\|y-1| + e(y-1)^2] \\
&\leq \frac{e}{2}[4(x-1)^2 + 2(x-1)^2 + 2(y-1)^2 + (y-1)^2] \leq 3e\big[(x-1)^2 + (y-1)^2\big]
\end{aligned}
$$